Hybrid Cellular Automata for Avascular Tumour Growth
Paper Analysis and Project Implementation

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Computational Models for Complex Systems

Talk structure

Part I - Paper (Gerlee & Anderson, 2007)

- Paper goal
- Modeling workflow
- Architecture and equations
- Main results

Part II - Our implementation

- Implementation in `model.py`
- ANN training and default-weight issue
- Multi-seed protocol in `sim_over_multiple_seed.ipynb`
- Results and differences vs paper

Paper: Gerlee & Anderson (2007)

Paper goal

Main objective

- Model avascular tumour growth with explicit clonal evolution.
- Study how oxygen availability changes morphology, growth, and phenotype selection.

Core modeling idea

- Each cell has an evolvable ANN-based response policy.
- Cell division introduces mutations in ANN parameters.
- Competition for resources creates eco-evolutionary selection.

Modeling workflow in the paper

Hybrid architecture

- Discrete layer: cellular automaton on a lattice (one cell per site).
- Continuous layer: reaction-diffusion fields (oxygen, glucose, hydrogen ions).
- Bidirectional coupling between CA decisions and PDE source/sink terms.

Simulation cycle

- Solve fields on the grid.
- For each cell: evaluate ANN, select action, apply rules.
- Update uptake/production and move to next step.

Dynamic model as a coupled system

Full paper formulation (conceptual system)

$$\begin{cases} \partial_t c = D_c \Delta c - f_c(s, R), \\ \partial_t g = D_g \Delta g - f_g(s, R), \\ \partial_t h = D_h \Delta h + f_h(s, R), \\ V_j = T\left(\sum_k w_{jk} x_k - \theta_j\right), \quad O_i = T\left(\sum_j W_{ij} V_j - \phi_i\right), \\ A_{ij}(t) = \arg\max\{O_P, O_Q, O_A\}, \\ s_{ij}(t+\Delta t) = \mathcal{U}\left(s_{ij}(t), A_{ij}(t), \mathcal{N}_{ij}(t), c, g, h\right), \\ \omega_{\text{child}} = \omega_{\text{parent}} + m \odot \xi, \;\; m_k \sim \text{Bernoulli}(p), \;\; \xi_k \sim \mathcal{N}(0, \sigma^2). \end{cases}$$

Metabolic modulation (paper Eq. 1)

$$F(R) = \max\left(k(R - T_r) + 1, \, 0.25\right).$$

Non-dimensionalization used in the paper

Change of variables

$$\tilde{x} = \frac{x}{L}, \qquad \tilde{t} = \frac{t}{\tau}, \qquad \tilde{c} = \frac{c}{c_0}, \qquad \tilde{g} = \frac{g}{g_0}, \qquad \tilde{h} = \frac{h}{h_0}.$$

Non-dimensional parameters

$$\tilde{D}_i = \frac{D_i \tau}{L^2}, \qquad \tilde{r}_c = \frac{\tau n_0 r_c}{c_0}, \qquad \tilde{r}_g = \frac{\tau n_0 r_g}{g_0}, \qquad \tilde{r}_h = \frac{\tau n_0 r_h}{h_0}.$$

- In the paper, simulations are run in these non-dimensional variables.
- In our project, parameters such as `c0`, `Dc`, `Dt`, `d`, `rc` are used in this normalized setting.

Paper simulation protocol and outputs

Protocol

- Grid size $N = 400$, initial seed of 4 central cells.
- Oxygen conditions: c_0, $c_0/2.5$, $c_0/10$.
- 20 stochastic runs per condition.
- Main paper plots focus on the reduced oxygen-driven subsystem.

Measured outputs

- Total tumour cells (alive + dead).
- Invasive distance from tumour center.
- Shannon diversity index:

$$H(t) = -\sum_k p_k(t) \log p_k(t).$$

Paper results: growth and morphology

Observed regimes

- $c_0/10$: early hypoxia and fingered morphology.
- $c_0/2.5$: delayed death onset and weaker fingering.
- c_0: rounder growth with necrotic core + viable rim.

Reference quantitative scale at $t = 100$ (paper discussion)

$$N_{c_0} \approx 5 \times 10^4, \qquad N_{c_0/10} \approx 1 \times 10^4.$$

Paper results: evolutionary dynamics

Shannon trend

- $c_0/10$: rapid increase up to $H \approx 0.9$, then slow decrease.
- c_0: slower increase, later convergence around $H \approx 0.8$.

Phenotype shift (Table 4 in paper)

- Initial response vector $(P, Q, A) = (0.67, 0.18, 0.15)$.
- Evolved at c_0: $(0.68, 0.17, 0.15)$, consumption ≈ 1.00.
- Evolved at $c_0/10$: $(0.83, 0.11, 0.06)$, consumption ≈ 1.14.

Our Project Implementation

How we implemented the model (`model.py`)

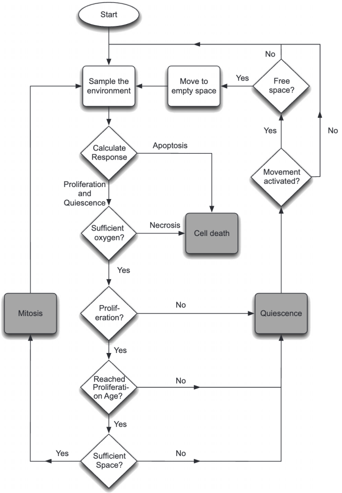
Implemented model scope

- Oxygen-only reduced model.
- States: `EMPTY`, `PROLIFERATING`, `QUIESCENT`, `NECROTIC`, `APOPTOTIC`.
- ANN input reduced to $\epsilon = [n_{\text{neighbors}}, c]$.

Numerical update in code

- ANN forward $\rightarrow$ action (P, Q, A)
- age update with $F(R)$
- apoptosis / necrosis / proliferation rules
- $c^{t+\Delta t} = c^t + \Delta t\left(D_c \Delta c - \alpha\right),\ c \geq 0$

Implementation diagram



Default parameters used in our implementation

Params defaults from `model.py`

Group	Values
Lattice/runtime	$N = 400$, `seed_radius`$= 4$, `steps`$= 200$, `rng_seed`$= 0$
Oxygen field	$c_0 = 1.0$, $D_c = 0.05$, $\Delta t = 0.2$, $d = 1.0$
Uptake/modulation	$r_c = 0.4$, $q = 5.0$, $T_r = 0.675$, $k = 6.0$
Thresholds	$c_{\text{apoptosis}} = 0.15$, $n_{\text{quiescence}} = 3.0$
Cell cycle	$\Delta t_{\text{age}} = 1.0$, $A_p = 1.0$, $A_{p,s} = 0.5$
Mutation	$p = 0.01$, $s = 0.25$

Note

These defaults are exactly the ones used when notebooks instantiate `p = Params()`.

ANN implementation and default-weight issue

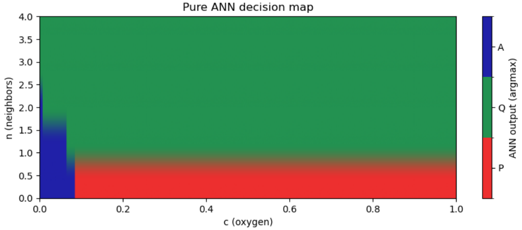
`ann_training.ipynb` pipeline

- Generate 30k samples from rule labels: A if low oxygen, else P if low crowding, else Q.
- Train ANN with mini-batch gradient descent and early stopping.
- Save trained parameters to `.env_ann_params`.

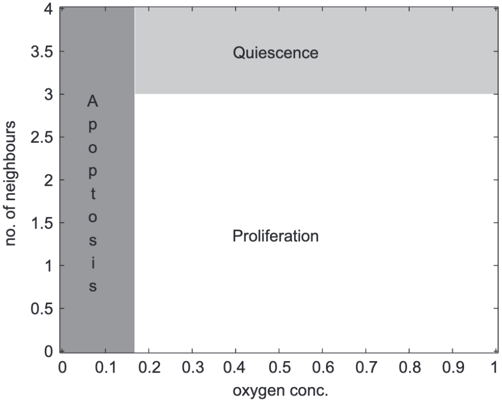
Observed issue and fix

- Default ANN (untrained) does not reproduce target policy: accuracy ≈ 0.5404, recall$_P \approx 0.334$, recall$_A \approx 0.200$.
- Trained ANN solves it: accuracy 0.9994, balanced accuracy 0.9987, macro F1 0.9991, ROC-AUC 1.0000.

ANN decision boundary before training

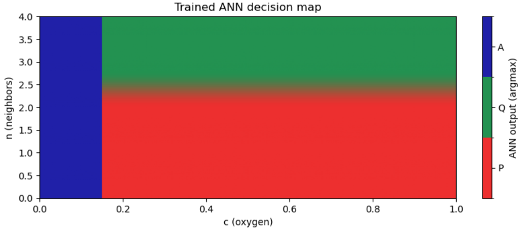


Our pre-training ANN map

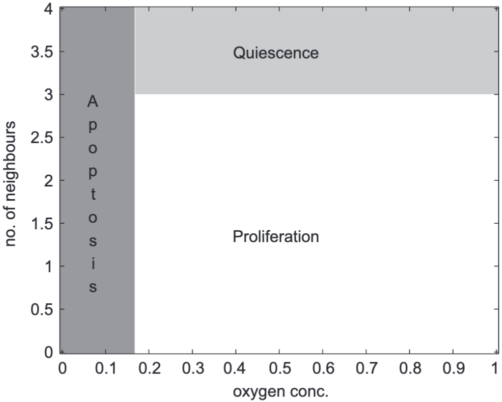


Reference decision boundary

ANN decision boundary after training



Our post-training ANN map



Reference decision boundary

What `sim_over_multiple_seed.ipynb` does

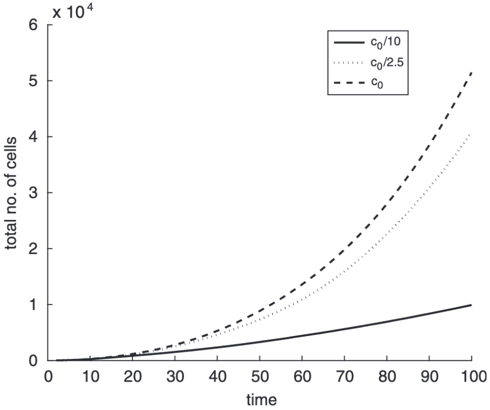
Multi-seed protocol

- Load trained ANN parameters.
- Simulate 20 seeds for each oxygen case: c_0, $c_0/2.5$, $c_0/10$.
- Aggregate trajectories as mean $\pm$ std.

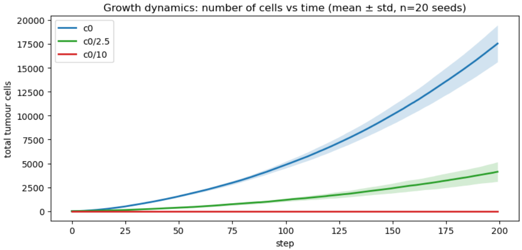
Collected outputs

- Total tumour cells over time.
- Invasive distance over time.
- Shannon diversity over time.
- Summary table at $t = 100$.

Total cells over time across oxygen levels

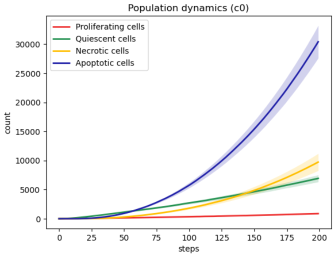


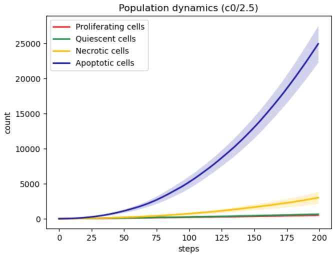
Reference

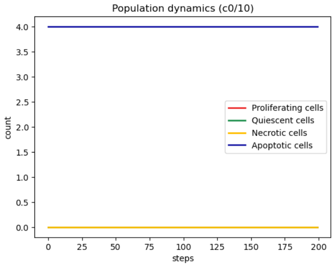


Our result

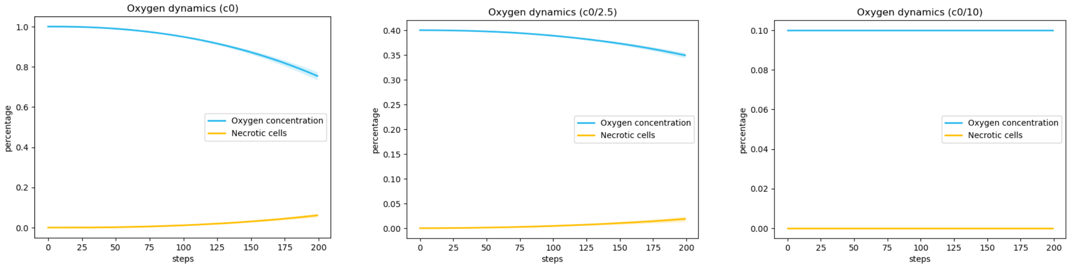
Population dynamics across oxygen cases







Oxygen and necrotic percentage across oxygen cases



Our results (20 seeds, summary at $t = 100$)

Tumour size (mean $\pm$ std)

Oxygen case	Mean tumour size	Std
c_0	4772.05	340.78
$c_0/2.5$	1158.40	159.04
$c_0/10$	0.00	0.00

Trend

Lower oxygen $\Rightarrow$ lower growth and stronger stress effects.

Differences vs paper results

What is consistent

- Correct qualitative ordering across oxygen conditions.
- Oxygen clearly controls growth/invasion behavior.

What differs quantitatively

- Our sizes are smaller (e.g., $\sim 4.8 \times 10^3$ vs paper $\sim 5 \times 10^4$ at $t = 100$, c_0 case).
- In our setup, $c_0/10$ collapses to zero by $t = 100$.

Main reasons

- Reduced oxygen-only subsystem and simplified ANN I/O.
- Different calibration of uptake, age, and effective time-scale.
- Stricter starvation dynamics in our step-wise update.

Conclusions

Paper side

- Hybrid CA-PDE framework explains oxygen-driven morphology and selection.

Implementation side

- Reproducible implementation with trained ANN and multi-seed analysis.
- Qualitative agreement achieved; quantitative calibration remains open.

References

- P. Gerlee, A.R.A. Anderson (2007), *An evolutionary hybrid cellular automaton model of solid tumour growth*, Journal of Theoretical Biology 246:583–603.
- Project files: `simulation/model.py`, `simulation/ann_training.ipynb`, `simulation/sim_over_multiple_seed.ipynb`.